

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	03 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection & Extraction from Database	Collect electricity consumption data, Import data from CSV/Excel/Database, Validate records, Store data in database
FR-2	Data Preparation	Remove duplicate records, Handle missing values, Clean inconsistent data, Format date/time, Create calculated fields
FR-3	Data Visualization	Create bar charts, Line charts, Pie charts, Monthly and yearly consumption graphs, Appliance-wise usage charts
FR-4	Dashboard	Display total electricity usage, Show electricity bill estimate, Filter by date/month/year, Interactive dashboard, Export dashboard
FR-5	Story	Generate insights from data, Highlight high-consumption periods, Compare monthly/yearly usage, Suggest energy-saving actions
FR-6	Performance Testing	Test dashboard loading speed, Verify report generation time, Validate data accuracy, Check system response time
FR-7	Web Intergration	User login, Connect dashboard to database, Publish dashboard on web, Responsive design for mobile and desktop

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface should be simple, easy to understand, and user-friendly.
NFR-2	Security	User data and electricity usage information should be securely stored and protected.
NFR-3	Reliability	The system should provide accurate electricity usage and bill calculations with minimal downtime.

NFR-4	Performance	The application should load quickly and display data within a few seconds.
NFR-5	Availability	The system should be available 24/7 with minimal downtime so users can access electricity consumption data anytime.
NFR-6	Scalability	The system should support an increasing number of users and larger volumes of electricity consumption data without performance issues.